

(a) Plugins description

YAML input deck

Ordered list of
ONIKA operators

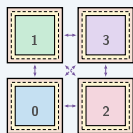
Onika runtime
Operator graph ·
Slot management
MPI+ OpenMP/CUDA

**External
plugins**
e.g. DEM-IBM-
LBM coupling



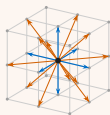
HippoLBM plugins

grid



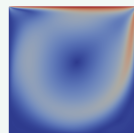
Data storage, fields,
LBM parameters;
block-based
load balancing

collision



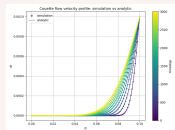
BGK / MRT collision
Streaming phase
Compute macroscopic
quantities ρ , $\mathbf{u}$

bcs



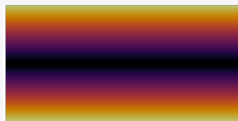
Bounce-back
Neumann
Lid-driven cavity

io



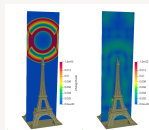
Print logs,
Para View files
Post process-
ing analysis

prepo



Field pre-
initialisation for
given flow regimes,
e.g. double Couette

obstacle



Fixed solids:
walls and
quadric-defined
geometries

(b) Operator sequence, coloured by plugin

